

Technical Summary

1. Technical Summary — Internal Notes for Future Pipeline Design

Phase 1 — Block 2

Initial Data Loading and Structural Inspection

- Dataset shape: 284,807 rows × 31 columns
- Features:
 - Time
 - Amount
 - 28 anonymized PCA-transformed variables (V1–V28)
- Target variable:
 - Class
 - 0 = non-fraud
 - 1 = fraud
- Data types:
 - almost all features are float64
 - target variable is int64
- Dataset memory usage:
 - approximately 67.36 MB
- Fraud distribution:
 - 284,315 non-fraud transactions
 - 492 fraud transactions
- Fraud rate:
 - approximately 0.17%
- Unique values
 - Time unique values ≈ 124,592
 - Amount unique values ≈ 32,767
- Initial evidence of
 - severe class imbalance
 - highly skewed target distribution
- PCA-transformed variables:
 - original banking variables were anonymized
 - features are already transformed and numerically standardized-like
 - feature semantics are unknown
- Time variable:
 - represents elapsed seconds since the first transaction in the dataset
 - not a real timestamp or calendar datetime
- Initial implications for future pipeline design:
 - imbalance handling will likely be critical
 - recall-oriented evaluation metrics may become important
 - feature scaling requirements must be evaluated
 - temporal feature engineering from Time may be explored
 - possible future engineered features:
 - transaction velocity
 - cyclical behavior

- temporal patterns
 - hour-like features
- Initial indications suggest the dataset is relatively clean structurally:
 - no immediate schema inconsistencies
 - all variables already numeric
 - suitable for ML pipelines and PySpark processing

Phase 1 — Block 3

Data Quality Assessment

- Dataset contains:
 - 0 missing values
 - 0 infinite values
- Dataset appears structurally clean and numerically stable.
- No immediate need for:
 - null imputation,
 - categorical encoding,
 - or invalid-value correction.

Duplicate analysis

- Detected duplicated rows:
 - 1,081 duplicated records
- Duplicate handling may later become part of preprocessing strategy.
- Duplicate analysis is important because duplicated fraud records may introduce:
 - leakage,
 - inflated metrics,
 - or unrealistic fraud representation.

Feature uniqueness and sparsity

- PCA-transformed variables (V1–V28) exhibit:
 - high cardinality,
 - continuous numerical distributions,
 - and no suspicious constant behavior.
- Time unique values:
 - 124,592
- Amount unique values:
 - 32,767
- No suspicious low-information or constant features were detected.

Zero-value analysis

- Amount = 0 appears in:
 - 1,825 transactions
- Zero-amount transactions may require later behavioral investigation.
- Presence of zero-value fraud transactions suggests:
 - zero-amount operations may still contain fraud-related signal.

Outlier inspection

- Outlier inspection revealed substantial extreme-value behavior across multiple variables.

Highest outlier percentages

- V27 outliers $\approx 13.75\%$
- Amount outliers $\approx 11.20\%$
- V28 outliers $\approx 10.65\%$
- V20 outliers $\approx 9.75\%$
- V8 outliers $\approx 8.47\%$

Outlier interpretation

- Fraud datasets naturally contain strong outlier behavior.
- Skewness and extreme values are expected characteristics of transactional financial data.
- Many extreme-value transactions are legitimate customer behavior.
- Some fraud cases may also exist within extreme-value regions.
- Aggressive outlier removal may therefore damage fraud detection performance.
- Outliers should be handled carefully rather than automatically removed.

Distribution and skewness analysis

- Strong skewness was identified across several variables.

Highest skewness magnitudes

- Amount skewness ≈ 16.98
- V28 skewness ≈ 11.19
- V8 skewness ≈ -8.52
- V23 skewness ≈ -5.88
- V2 skewness ≈ -4.62

Distribution interpretation

- In transactional fraud datasets:
 - skewness in Amount is expected and normal,
 - many outliers are legitimate transactional behavior,
 - but some fraud cases may also exist within extreme-value regions.
- Amount may later require:
 - scaling experiments,
 - logarithmic transformations,
 - or robust preprocessing strategies.
- Time distribution appears relatively symmetric compared to other variables.

Initial preprocessing implications

- Likely preprocessing considerations:
 - evaluate duplicate removal strategy,
 - preserve potentially informative outliers,
 - test scaling approaches,
 - investigate imbalance-aware preprocessing methods.
- Likely unnecessary preprocessing:
 - missing-value imputation,
 - infinite-value correction,
 - categorical encoding.

Additional statistical diagnostics

- Additional statistical diagnostics were incorporated for:
 - outlier inspection,
 - skewness analysis,
 - and distribution characterization,
 - improving future production pipeline decisions.

Phase 1 — Block 4

Target Variable and Class Imbalance Analysis

- Target variable:
 - Class
 - 0 = non-fraud
 - 1 = fraud
- Class distribution:
 - 284,315 non-fraud transactions
 - 492 fraud transactions
- Fraud percentage:
 - approximately 0.1727%
- Non-fraud percentage:
 - approximately 99.8273%
- Imbalance ratio:
 - approximately 577.88 : 1
- Dataset exhibits:
 - extreme class imbalance,
 - highly skewed target distribution,
 - and severe minority-class scarcity.
- Standard accuracy metrics will likely become misleading for this problem.
- High overall accuracy may still correspond to poor fraud detection performance.
- Future evaluation should prioritize:
 - Recall,

- Precision,
 - F1-score,
 - ROC-AUC,
 - PR-AUC,
 - confusion matrix analysis,
rather than accuracy alone.
- Fraud detection modeling will likely require imbalance-aware strategies:
 - class weighting,
 - undersampling,
 - oversampling,
 - threshold tuning,
 - or imbalance-sensitive algorithms.
- Stratified train/test splitting will likely become necessary to preserve fraud distribution consistency.
- Threshold optimization may become more important than default classification thresholds.
- False negatives will likely represent a higher business risk than false positives.
- Visual inspection confirmed that fraud transactions represent only a tiny fraction of total observations.
- The imbalance severity strongly suggests that:
 - baseline models may initially struggle with recall,
 - minority-class learning stability may become challenging,
 - and model evaluation variance may be significant.
- These findings will later influence:
 - train_model.py
 - evaluate_model.py
 - imbalance handling pipeline design
 - threshold **optimization strategy**.
 - and final production model selection.

Phase 1 — Block 5

Exploratory Data Analysis (EDA)

- Transaction amounts exhibit:
 - extreme right skewness,
 - heavy-tail behavior,
 - and substantial outlier presence.
- Global Amount statistics:
 - mean ≈ 88.35
 - median ≈ 22.00
 - standard deviation ≈ 250.12
 - maximum transaction $\approx 25,691.16$
- Most transactions are concentrated at relatively low monetary values.

- Raw Amount distribution is heavily compressed near zero.
- Log transformation ($\log(\text{Amount} + 1)$) significantly improves distribution interpretability.
- Log-transformed Amount distributions reveal:
 - clearer density structures,
 - better visualization of transactional behavior,
 - and more visible differences between fraud and non-fraud classes.
- Fraud and non-fraud Amount distributions overlap substantially.
- Amount alone is insufficient to reliably separate fraud from non-fraud transactions.
- Large transaction amounts are not inherently fraudulent.
- Many extreme-value transactions are legitimate non-fraud behavior.
- Some fraud-related behavioral differences appear in lower-to-mid transaction ranges.
- Fraud distributions appear:
 - smoother,
 - more dispersed,
 - and behaviorally different from non-fraud distributions after transformation.
- Findings suggest:
 - transformed Amount features may be more useful than raw Amount values,
 - especially for scaling-sensitive algorithms.

Amount comparison by class

Non-fraud transactions (Class = 0)

- mean ≈ 88.29
- median ≈ 22.00
- standard deviation ≈ 250.11
- maximum $\approx 25,691.16$

Fraud transactions (Class = 1)

- mean ≈ 122.21
- median ≈ 9.25
- standard deviation ≈ 256.68
- maximum $\approx 2,125.87$

Distributional interpretation

- Fraud transactions exhibit:
 - lower median transaction amounts,
 - but higher mean transaction amounts.
- This suggests:
 - many fraud transactions occur at relatively low values,
 - while a smaller subset involves moderately high amounts.
- Fraud behavior therefore appears:
 - asymmetric,
 - dispersed,
 - and not easily separable through simple amount thresholds.

Zero-amount transaction analysis

- 1,825 transactions have Amount = 0
- among them:
 - 1,798 are non-fraud
 - 27 are fraud
- Presence of fraud within zero-amount transactions suggests:
 - zero-value operations may still contain useful fraud signal,
 - and should not automatically be discarded during preprocessing.

High-value transaction analysis

- 99th percentile threshold ≈ 1017.97
- 2,849 transactions exceed this threshold
- among them:
 - 2,840 are non-fraud
 - 9 are fraud
- Fraud is not concentrated exclusively among high-value transactions.
- High transaction amount alone appears insufficient for reliable fraud identification.

Time analysis

- Time statistics:
 - mean $\approx 94,813.86$
 - median $\approx 84,692$
 - maximum $\approx 172,792$
- Transaction activity over Time exhibits cyclical behavior patterns.
- Fraud activity does not appear uniformly distributed across time.
- Fraud density changes across temporal regions differently from non-fraud behavior.
- Temporal behavior may contain nonlinear predictive signal despite weak linear correlation.

Correlation analysis

- Correlation analysis showed:
 - Time $\leftrightarrow$ Class ≈ -0.0123
 - Amount $\leftrightarrow$ Class ≈ 0.0056
- Extremely weak linear correlations indicate:
 - fraud signal is likely nonlinear,
 - interaction-based,
 - and distributional rather than linearly separable.
- Weak correlation does not imply lack of predictive value.
- Fraud-related signal likely exists through:
 - nonlinear relationships,
 - multivariable interactions,
 - and distributional behavior patterns.

Modeling implications

- High overlap between fraud and non-fraud distributions suggests:
 - threshold-based rules alone will likely perform poorly.
- Future modeling will likely require:
 - nonlinear models,
 - interaction-aware algorithms,
 - or ensemble-based approaches.
- Initial feature engineering implications:
 - evaluate log-transformed Amount features,
 - investigate temporal feature extraction,
 - preserve informative outliers,
 - and test scaling-sensitive preprocessing strategies.

Additional exploratory diagnostics

- Additional fraud-vs-nonfraud density visualizations were incorporated to:
 - compare population-level distributions,
 - identify subtle behavioral shifts,
 - and evaluate distribution overlap between classes.

Phase 1 — Block 6

Correlation and Relationship Analysis

- Correlation analysis was performed against:
 - target variable Class
 - where:
 - 0 = non-fraud
 - 1 = fraud
- Positive correlations therefore indicate:
 - features whose values tend to increase in fraudulent transactions.
- Negative correlations indicate:
 - features whose values tend to decrease in fraudulent transactions.

Strongest positive correlations with fraud (Class = 1)

- V11 $\approx$ 0.1549
- V4 $\approx$ 0.1334
- V2 $\approx$ 0.0913
- V21 $\approx$ 0.0404

Strongest negative correlations with fraud (Class = 1)

- V17 $\approx$ -0.3265
- V14 $\approx$ -0.3025
- V12 $\approx$ -0.2606
- V10 $\approx$ -0.2169
- V16 $\approx$ -0.1965

- $V3 \approx -0.1930$
- $V7 \approx -0.1873$

Correlation interpretation

- Negative fraud correlations exhibit larger magnitudes than positive correlations.
- Fraud behavior in this dataset appears more associated with:
 - reductions in certain latent PCA dimensions, rather than increases alone.
- No individual variable exhibits extremely high linear correlation with fraud.
- Fraud detection in this dataset is therefore:
 - not linearly separable,
 - multivariable,
 - and behaviorally complex.

Amount and Time relationships

- Amount $\leftrightarrow$ Class ≈ 0.0056
- Time $\leftrightarrow$ Class ≈ -0.0123
- Both variables exhibit extremely weak linear relationships with fraud.
- Weak linear correlation does not imply lack of predictive value.
- Earlier EDA findings suggest:
 - nonlinear relationships,
 - interaction effects,
 - and distributional behavior may still contain predictive signal.

Distribution comparison analysis

- KDE distribution comparisons were performed for:
 - V11
 - V4
 - V17
 - V14

Positive-correlation feature behavior

- V11 and V4 distributions show:
 - fraud transactions shifted toward higher feature values,
 - while non-fraud transactions remain concentrated near lower regions.
- These variables exhibit:
 - partial class separability,
 - suggesting useful predictive signal.

Negative-correlation feature behavior

- V17 and V14 distributions show:
 - fraud transactions shifted strongly toward negative values,
 - while non-fraud transactions cluster tightly near zero.

- These variables exhibit:
 - the strongest visible class-separation patterns observed so far.
- Fraud and non-fraud transactions occupy:
 - substantially different latent feature regions within these variables.

Relationship structure observations

- Several PCA-transformed variables contain:
 - meaningful fraud-separation structure,
 - despite severe class imbalance.
- Correlation heatmap analysis confirmed:
 - limited multicollinearity among selected variables.
- Most feature-to-feature correlations remain near zero.
- PCA transformation appears effective at:
 - producing relatively orthogonal latent variables.

Leakage inspection

- Leakage inspection searched for:
 - absolute correlations greater than 0.9
- No suspicious features were detected.
- Only:
 - Class $\leftrightarrow$ Class = 1.0 appeared, which is expected.
- No evidence of obvious target leakage was identified.

Modeling implications

- Fraud signal appears distributed across multiple latent variables.
- Several PCA-transformed variables are likely to become:
 - highly informative predictive features.
- Partial separability patterns suggest:
 - machine learning models should be capable of learning meaningful fraud behavior.
- Because separability remains incomplete:
 - simple threshold rules will likely perform poorly.
- Future modeling will likely benefit from:
 - nonlinear algorithms,
 - ensemble methods,
 - interaction-aware models,
 - and probability-based decision boundaries.
- Potential future considerations:
 - feature importance analysis,
 - SHAP interpretation,
 - regularization experiments,
 - and latent-space visualization techniques.

Phase 1 — Block 7

Exploratory Feature Engineering

Amount transformation experiments

- Original Amount distribution exhibits:
 - extreme right skewness,
 - heavy-tail behavior,
 - and strong outlier influence.
- Original Amount statistics:
 - mean ≈ 88.35
 - median ≈ 22.00
 - standard deviation ≈ 250.12
 - maximum $\approx 25,691.16$
- Large separation between:
 - mean and median,
 - and extremely large maximum values,indicates severe skewness and variance instability.

Log transformation analysis

- Applied transformation:

$\log_{10}(\text{Amount}+1)$

- Log-transformed statistics:
 - mean ≈ 3.15
 - median ≈ 3.14
 - standard deviation ≈ 1.66
 - maximum ≈ 10.15
- Log transformation substantially reduced:
 - skewness,
 - variance dominance,
 - and outlier magnitude effects.
- Mean and median became substantially closer after transformation, indicating improved distribution stability.
- Transformed distribution remains non-normal but becomes:
 - smoother,
 - more interpretable,
 - and more model-friendly.
- Log transformation revealed hidden density structure that was compressed in the raw distribution.
- Findings suggest:
 - transformed Amount features may provide better learning behavior than raw Amount values.

Scaling experiments

- Three scaling approaches were evaluated:
 - StandardScaler
 - RobustScaler

- MinMaxScaler

StandardScaler observations

- StandardScaler successfully centered data around zero.
- However:
 - skewness,
 - heavy tails,
 - and extreme-value influence remained substantial.
- Outliers continue to distort variance-based scaling behavior.

RobustScaler observations

- RobustScaler demonstrated:
 - improved resistance to outliers,
 - more stable central scaling behavior,
 - and reduced sensitivity to extreme values.
- Distribution remained skewed but appeared more stable than StandardScaler output.
- RobustScaler appears potentially more suitable for heavy-tailed transactional fraud data.

MinMaxScaler observations

- MinMaxScaler compressed nearly all observations close to zero because:
 - extreme outliers dominate scaling boundaries.
- Significant loss of visible feature resolution occurred among ordinary transactions.
- MinMax scaling may be less suitable without prior outlier treatment.

Imbalance experiment — undersampling

- Original class distribution:
 - non-fraud = 284,315
 - fraud = 492
- Original imbalance ratio:
 - approximately 578:1
- Exploratory undersampling created:
 - 492 fraud transactions
 - 492 non-fraud transactions
- Undersampling successfully produced a balanced dataset but removed:
 - more than 283,000 legitimate transactions.
- Undersampling may:
 - improve minority-class learning,
 - reduce majority-class dominance,
 - and improve fraud recall.
- However:
 - severe information loss may reduce generalization capability and legitimate-behavior representation.

- Findings suggest:
 - imbalance handling strategy should later compare:
 - undersampling,
 - class weighting,
 - oversampling,
 - SMOTE,
 - and threshold optimization.

Fraud vs non-fraud transformed distributions

- Log-transformed Amount distributions reveal:
 - clearer behavioral differences between fraud and non-fraud transactions.
- Fraud distribution appears:
 - broader,
 - smoother,
 - and behaviorally different from non-fraud density patterns.
- Non-fraud transactions remain:
 - more concentrated,
 - and exhibit sharper density peaks.
- Transformation improved visibility of latent transactional structure previously compressed in raw Amount distributions.

Modeling implications

- Preprocessing decisions are likely to strongly influence:
 - optimization stability,
 - feature usability,
 - minority-class learning,
 - and overall fraud detection performance.
- Future preprocessing pipeline should likely evaluate:
 - log-transformed Amount features,
 - robust scaling approaches,
 - imbalance-aware training methods,
 - and preprocessing consistency across train/test splits.
- Fraud detection remains:
 - highly overlapping,
 - nonlinear,
 - and behaviorally complex,
 even after exploratory preprocessing transformations.

Phase 1 — Block 8

Baseline Modeling

Dataset split configuration

- Dataset was divided using:
 - 80% training
 - 20% testing

- Stratified splitting was applied:
 - preserving the original fraud ratio in both subsets.
 - Test set composition:
 - non-fraud = 56,864
 - fraud = 98
 - Severe imbalance remained present during evaluation.
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Logistic Regression baseline

Configuration

- Logistic Regression was trained using:
 - class_weight="balanced"
 - max_iter=1000
- Log-transformed Amount feature was used instead of raw Amount.

Convergence behavior

- Training generated:
 - ConvergenceWarning
- Model still executed successfully.
- Warning indicates:
 - optimization reached iteration limit before full convergence.
- Possible causes:
 - severe imbalance,
 - complex feature geometry,
 - incomplete scaling consistency,
 - nonlinear fraud structure.
- Future pipeline may require:
 - additional scaling,
 - alternative solvers,
 - or higher iteration limits.

Logistic Regression performance

- Accuracy ≈ 0.9746
- Precision ≈ 0.0588
- Recall ≈ 0.9184
- F1-score ≈ 0.1106
- ROC-AUC ≈ 0.9718

Actual / Predicted	Non-Fraud	Fraud
Non-Fraud	55,424	1,440
Fraud	8	90

Logistic Regression behavior analysis

- Model achieved:
 - extremely high fraud recall (~92%)
- Very few fraud cases were missed:
 - only 8 false negatives.
- However:
 - precision remained extremely poor (~5.9%) because:
 - large numbers of legitimate transactions were incorrectly flagged as fraud (1,440 false positives).
- Model strongly prioritized:
 - fraud detection sensitivity,
 - over:
 - precision stability.
- This behavior is common in:
 - imbalance-sensitive linear models using balanced class weighting.

Interpretation

- Logistic Regression demonstrates:
 - fraud signal is learnable,
 - but linear decision boundaries are insufficient for clean class separation.
 - High ROC-AUC suggests:
 - probability ranking remains strong,
 - even though classification threshold performance is poor.
-

Random Forest baseline

Configuration

- Random Forest trained using:
 - 100 trees
 - class_weight="balanced"

Random Forest performance

- Accuracy ≈ 0.9995
- Precision ≈ 0.9605
- Recall ≈ 0.7449
- F1-score ≈ 0.8391
- ROC-AUC ≈ 0.9529

Actual / Predicted	Non-Fraud	Fraud
Non-Fraud	56,861	3
Fraud	25	73

Random Forest behavior analysis

- Random Forest dramatically reduced false positives:
 - only 3 legitimate transactions incorrectly flagged.
- Fraud precision became extremely high (~96%).
- However:
 - fraud recall decreased compared to Logistic Regression.
- Model missed:
 - 25 fraud cases.
- Random Forest therefore prioritized:
 - prediction confidence and precision, over:
 - maximum fraud capture.

Interpretation

- Results strongly suggest:
 - nonlinear relationships exist within the fraud patterns.
- Tree-based ensemble learning appears substantially better suited than linear modeling for this dataset.
- Random Forest achieved:
 - significantly stronger balance between precision and recall.

Comparative modeling insights

Logistic Regression vs Random Forest:

Metric	Logistic Regression	Random Forest
Precision	Very low	Very high
Recall	Very high	Moderate-high
False Positives	Very high	Extremely low
False Negatives	Very low	Higher
Interpretability	High	Moderate
Nonlinear Learning	Weak	Strong

Important strategic insight

- Different models optimize different business tradeoffs:
 - Logistic Regression prioritizes fraud capture,
 - Random Forest prioritizes fraud certainty.
- Final production decisions will likely require:
 - threshold optimization,
 - business-cost analysis,
 - and recall-vs-precision tradeoff tuning.

Feature importance analysis

Top Random Forest feature importances:

Feature	Importance
V14	0.1799
V10	0.1154
V12	0.0962
V4	0.0956
V17	0.0951

Importance interpretation

- Feature importance measures:
 - relative contribution to impurity reduction across the forest.
- Importance values are:
 - normalized contributions,
 - not probabilities or predictive metrics.
- Findings strongly align with:
 - earlier correlation analysis,
 - KDE distribution separation,
 - and latent PCA feature behavior observed in prior blocks.

Consistency across analyses

- Variables repeatedly identified as important:
 - V14
 - V17
 - V12
 - V10
 - V4

- Strong consistency across:
 - correlation analysis,
 - distribution comparisons,
 - and model-based importance suggests:
 - these latent variables contain substantial fraud signal.
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Modeling implications

Fraud learnability

- Results provide strong evidence that:
 - the dataset contains meaningful learnable fraud structure.

Nonlinearity

- Random Forest significantly outperformed Logistic Regression in overall balance.
- Fraud detection problem appears:
 - nonlinear,
 - interaction-heavy,
 - and poorly represented by simple linear boundaries.

Future modeling directions

Future phases should likely evaluate:

- XGBoost
- LightGBM
- CatBoost
- calibrated thresholds
- probability optimization
- precision-recall tuning
- ensemble strategies
- advanced imbalance handling
- SHAP explainability
- feature interaction analysis

Evaluation methodology implications

- Accuracy alone is not reliable for this dataset.
- Future evaluation should prioritize:
 - Recall
 - Precision
 - F1-score
 - ROC-AUC
 - Precision-Recall AUC

- business-cost-aware metrics

Phase 1 — Block 9

Model Evaluation and Metric Comparison

Evaluation focus

- This block focused on:
 - imbalance-aware evaluation,
 - operational fraud-detection tradeoffs,
 - threshold-independent metrics,
 - and business-oriented model interpretation.
- Evaluation emphasized:
 - Precision
 - Recall
 - F1-score
 - ROC-AUC
 - PR-AUC
 - False Positive Rates
 - False Negative Rates

ROC-AUC analysis

Model	ROC-AUC
Logistic Regression	0.9718
Random Forest	0.9529

Interpretation

- Both models achieved extremely strong ROC-AUC values.
- Results confirm:
 - substantial learnable fraud structure exists in the dataset.
- Logistic Regression achieved slightly higher ROC-AUC despite weaker operational precision.
- This suggests:
 - Logistic Regression probability ranking remains strong, even though:
 - default threshold classification behavior is suboptimal.
- Findings indicate:

- future threshold tuning may substantially improve linear-model performance.

PR-AUC analysis

Model	PR-AUC
Logistic Regression	0.7100
Random Forest	0.8543

Interpretation

- PR-AUC proved substantially more informative than ROC-AUC under severe class imbalance.
- Random Forest significantly outperformed Logistic Regression in PR-AUC.
- Results suggest:
 - Random Forest generates operationally stronger fraud alerts, particularly under imbalance-sensitive evaluation conditions.
- Findings reinforce:
 - PR-AUC should become a primary evaluation metric in future fraud-model experiments.

Precision vs recall tradeoff

Logistic Regression

Metric Value

Precision 0.0588

Recall 0.9184

Interpretation

- Logistic Regression strongly prioritized:
 - fraud capture sensitivity.
- Model detected:
 - approximately 92% of fraud cases.
- However:
 - precision remained extremely poor.
- Large numbers of legitimate transactions were incorrectly classified as fraud.

Operational implications

- High fraud recall may reduce:
 - undetected fraud loss.
- Extremely low precision may create:
 - analyst overload,
 - investigation inefficiency,
 - customer friction,
 - and excessive operational cost.

Random Forest

Metric	Value
Precision	0.9605
Recall	0.7449

Interpretation

- Random Forest prioritized:
 - fraud prediction confidence and precision.
- Fraud alerts became highly reliable.
- However:
 - fraud recall decreased relative to Logistic Regression.
- Model became substantially more conservative.

Operational implications

- Very low false-positive generation improves:
 - analyst efficiency,
 - operational stability,
 - and customer experience.
 - Increased false negatives may still permit:
 - meaningful fraud leakage.
-

False positive analysis

Model	False Positives
Logistic Regression	1,440
Random Forest	3

False Positive Rates:

Model	FPR
Logistic Regression	0.0253
Random Forest	0.000053

Interpretation

- Random Forest dramatically reduced false-positive generation.
- Logistic Regression produced extremely noisy fraud alerts.
- False positives represent:
 - legitimate customers incorrectly flagged as fraud.
- Operational consequences may include:
 - blocked transactions,
 - manual investigations,
 - analyst fatigue,
 - and customer dissatisfaction.

Important insight

- False positive management appears likely to become one of the most important optimization dimensions later in the project.

False negative analysis

Model	False Negatives
Logistic Regression	8
Random Forest	25

False Negative Rates:

Model	FNR
Logistic Regression	0.0816
Random Forest	0.2551

Interpretation

- Logistic Regression missed relatively few fraud cases.
 - Random Forest missed substantially more fraud.
 - False negatives represent:
 - fraud cases escaping detection.
 - Financial-loss exposure therefore remains:
 - an important operational tradeoff.
-

Imbalance effects

Accuracy limitations

Model	Accuracy
Logistic Regression	0.9746
Random Forest	0.9995

Interpretation

- Accuracy alone proved misleading under severe imbalance.
- Logistic Regression achieved:
 - very high accuracy,
 - while simultaneously:
 - generating extremely poor fraud precision.
- Findings confirm:
 - imbalance-aware metrics are essential in fraud-detection systems.

Most informative metrics identified

Future evaluation should prioritize:

- PR-AUC
- Recall

- Precision
 - F1-score
 - False Positive Rate
 - False Negative Rate
 - Precision-Recall tradeoffs instead of:
 - accuracy alone.
-

Threshold behavior observations

- Current models used default classification thresholds: [0.5]
- Results therefore do not represent:
 - fully optimized operational performance.
- Strong ROC-AUC values suggest:
 - threshold tuning may later improve:
 - precision,
 - recall,
 - and operational balance.

Future optimization directions

Potential future work:

- probability calibration
 - threshold optimization
 - business-cost-aware tuning
 - recall-vs-precision balancing
 - dynamic fraud thresholds
-

Model suitability conclusions

Logistic Regression

Strengths:

- high fraud sensitivity
- strong ranking capability
- interpretability
- simplicity

Weaknesses:

- massive false positive burden
- weak nonlinear representation
- operational instability

Potential future roles:

- risk scoring
 - secondary fraud screening
 - interpretable baseline benchmarking
-

Random Forest

Strengths:

- extremely high precision
- very low false positive rate
- strong nonlinear learning capability
- better operational balance

Weaknesses:

- lower fraud recall
- reduced interpretability
- more computational complexity

Potential future roles:

- production fraud detection
 - operational alert systems
 - ensemble-based fraud pipelines
-

Overall modeling conclusions

Most important findings

- Fraud detection performance is highly dependent on:
 - metric selection,
 - imbalance handling,
 - and business tradeoff priorities.
- Different models optimize:

- fundamentally different operational objectives.
- Fraud behavior appears:
 - nonlinear,
 - interaction-heavy,
 - and operationally sensitive to threshold decisions.
- Random Forest currently appears:
 - substantially more production-ready than Logistic Regression under default evaluation conditions.

Phase 1 — Block 10

Feature Importance and Model Insights

Feature importance analysis

- Random Forest feature importance identified a relatively small subset of latent variables as dominant fraud predictors.

Top Random Forest feature importances

Feature	Importance
V14	0.1799
V10	0.1154
V12	0.0962
V4	0.0956
V17	0.0951

Interpretation

- These variables contributed most strongly to:
 - impurity reduction,
 - decision boundaries,
 - and fraud-separation behavior across the ensemble.
- Findings strongly reinforce previous:
 - correlation analysis,
 - KDE distribution separation,
 - and model evaluation results.

Permutation importance analysis

Top permutation importances:

Feature	Permutation Importance
V12	0.6749
V14	0.5946
V11	0.3220
V3	0.3080
V4	0.2958

Interpretation

- Permutation importance measures:
 - degradation in model performance when a feature is randomly shuffled.
- Results indicate:
 - model performance depends heavily on a small subset of latent dimensions.
- V12 produced the largest performance degradation after permutation, suggesting:
 - extremely high operational predictive dependency.

Important insight

- Strong agreement between:
 - Random Forest importance,
 - permutation importance,
 - correlation analysis,
 - and KDE separationprovides strong evidence that:
 - fraud signal is stable and non-random.
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Feature consistency analysis

Common important features

{'V4', 'V3', 'V14', 'V17', 'V11', 'V12', 'V16', 'V10'}

Interpretation

- Multiple independent analytical methods repeatedly identified the same variables.
- This consistency significantly increases confidence that:
 - identified fraud indicators are genuine and structurally meaningful.
- Fraud-driving latent dimensions appear:
 - stable,
 - reproducible,
 - and highly predictive.

Correlation vs importance analysis

Observed behavior

- Several variables exhibited:
 - both strong correlation
 - and strong feature importance.

Examples:

Feature	Correlation	Importance
V14	-0.3025	0.1799
V17	-0.3265	0.0951
V4	0.1334	0.0956

Interpretation

- Features such as:
 - V14,
 - V17,
 - V12,
 - V10show:
 - strong linear fraud relationships
 - and strong nonlinear predictive utility.
- Other variables exhibited:
 - moderate correlation
 - but meaningful importance,suggesting:
 - nonlinear interactions,
 - conditional behavior,
 - and latent feature dependencies.

Important conclusion

- Fraud detection behavior appears:
 - strongly nonlinear,
 - interaction-driven,
 - and multidimensional.

KDE distribution analysis

Main observations

- Fraud and non-fraud distributions occupy:

- substantially different latent regions.
- Legitimate transactions remain:
 - tightly concentrated near central latent regions.
- Fraud transactions exhibit:
 - broader spread,
 - stronger deviation,
 - and abnormal latent displacement.

V14, V10, V12 behavior

- Fraud transactions shift strongly toward:
 - large negative latent values.
- Non-fraud observations remain concentrated near:

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V4 behavior

- Fraud transactions shift toward:
 - larger positive values.

Interpretation

- Different latent PCA dimensions capture:
 - different forms of anomalous transactional behavior.
- Fraud does not appear as:
 - a single behavioral pattern,
 - but rather:
 - multiple latent abnormality modes.

Low-importance feature analysis

Lowest-importance variables:

Feature	Importance
V13	0.0087
V5	0.0077
V28	0.0071
Time	0.0053
V24	0.0043

Interpretation

- These variables contributed minimally to fraud separation.

- Possible explanations:
 - weak standalone signal,
 - redundancy with stronger variables,
 - or limited fraud-specific structure.

Important observations

- Time remained consistently weak across:
 - correlation analysis,
 - feature importance,
 - and class-separation analysis.
- Log_Amount importance also remained relatively low:

0.01120.01120.0112

Important implication

- Fraud signal appears driven more strongly by:
 - latent behavioral structure
 - than:
 - simple transaction magnitude or timing alone.

Latent feature behavior

PCA-related interpretation

- Although original variables remain anonymized, PCA-transformed latent dimensions clearly preserve:
 - meaningful fraud-related structure.
- Fraud transactions systematically occupy:
 - abnormal latent regions,
 - particularly:
 - extreme negative latent regions.

Important insight

- Fraud behavior resembles:
 - multidimensional latent anomaly behavior,
 - rather than:
 - simple rule-based outlier behavior.

Nonlinear modeling implications

Evidence of nonlinearity

- Multiple variables exhibited:
 - meaningful predictive importance despite:
 - only moderate linear correlation.
- Results strongly support:
 - nonlinear feature interactions,
 - conditional fraud boundaries,
 - and ensemble-learning advantages.

Modeling implications

Future phases should likely prioritize:

- Random Forest ensembles
 - XGBoost
 - LightGBM
 - CatBoost
 - interaction-aware models
 - explainability frameworks
 - SHAP analysis
 - permutation-based interpretation
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Explainability and production implications

Important findings

- Fraud indicators are now:
 - reproducible,
 - interpretable,
 - and stable across methods.
- Strong explainability foundations now exist for:
 - future production reporting,
 - monitoring,
 - model governance,
 - and fraud-investigation workflows.

Potential future directions

- feature selection
- dimensionality reduction
- SHAP value analysis
- interaction visualization
- model simplification
- production monitoring of dominant latent variables

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PHASE 1 — FINAL CONCLUSIONS

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1. DATA UNDERSTANDING

- The dataset contains highly imbalanced fraud data:
Non-fraud transactions: 284,315
Fraud transactions: 492
- Fraud ratio:
approximately 0.1727%
- Fraud detection therefore represents a severe
class imbalance problem.
- The dataset contains anonymized PCA-transformed
variables (V1–V28), meaning:
fraud signal exists in latent behavioral dimensions.
- Transaction Amount and Time variables remain
interpretable original features.

2. DATA QUALITY ASSESSMENT

- No missing values detected.
- No infinite values detected.
- Duplicate rows detected:
1,081 rows
- Large numbers of outliers exist across multiple features.
- Heavy skewness observed especially in:
Amount,
V28,
V8,
V23.
- Skewness and outliers appear partially expected in:
transactional financial behavior.
- Extreme values should be handled carefully rather
than automatically removed because:
some fraud signal may exist within extreme regions.

3. EXPLORATORY DATA ANALYSIS

- Fraud and non-fraud transactions exhibit
partially separable behavioral structure.

- Amount distribution is highly right-skewed.
- Fraud transactions tend to occupy abnormal latent PCA regions.
- Zero-amount transactions and high-value transactions contain both:
legitimate behavior,
and fraud behavior.
- Fraud does not appear to follow:
a single simple transactional pattern.

4. FEATURE ENGINEERING INSIGHTS

- Log transformation substantially stabilized the Amount distribution.
- RobustScaler appeared more resistant to outlier distortion than:
StandardScaler,
and MinMaxScaler.
- Severe imbalance experiments demonstrated:
undersampling improves balance
but introduces major information loss.
- Future preprocessing pipelines should evaluate:
imbalance-aware strategies,
scaling consistency,
and transformation stability.

5. FEATURE RELATIONSHIP INSIGHTS

- Multiple latent variables consistently emerged as strong fraud indicators:
V14,
V12,
V10,
V17,
V4,
V11.
- Strong agreement exists across:
correlation analysis,
Random Forest importance,
permutation importance,
and KDE separation analysis.
- Fraud signal appears:
stable,
nonlinear,
and interaction-driven.

- Amount and Time alone provide limited predictive power relative to latent dimensions.

6. BASELINE MODELING INSIGHTS

- Logistic Regression achieved:
very high recall,
but extremely poor precision.
- Random Forest achieved:
very high precision,
dramatically lower false positives,
and stronger operational balance.
- Fraud detection performance depends heavily on:
metric selection,
threshold behavior,
and imbalance-aware evaluation.
- PR-AUC proved more informative than:
accuracy alone,
and ROC-AUC alone.
- Fraud detection appears strongly nonlinear,
suggesting:
advanced ensemble models may perform well.

7. OPERATIONAL FRAUD-DETECTION INSIGHTS

- Models with high recall detect more fraud cases
but may generate large numbers of false alerts.
- Models with high precision produce more reliable
fraud alerts but may miss some fraudulent activity.
- Fraud detection therefore requires balancing:
fraud capture,
false alerts,
customer experience,
and operational workload.
- Future phases will likely require threshold tuning
to optimize this balance for production scenarios.

8. TRANSITION TO PRODUCTION PIPELINE

Phase 1 established:

- reliable exploratory understanding,
- stable fraud indicators,
- preprocessing direction,
- baseline model behavior,
- and evaluation methodology.

The next phases will focus on:

- scalable preprocessing pipelines,
- production-oriented feature engineering,
- advanced model training,
- threshold optimization,
- explainability frameworks,
- and deployment-oriented architecture.

Future implementation may transition toward:

- PySpark-based scalable pipelines,
- distributed preprocessing,
- modular ML architecture,
- experiment tracking,
- and production monitoring systems.

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PHASE 1 COMPLETED

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Phase 1 successfully established:

- dataset understanding,
- fraud behavior characterization,
- preprocessing foundations,
- baseline modeling benchmarks,
- evaluation methodology,
- and interpretable fraud indicators.

The project now has:

- a strong analytical foundation,
- reproducible exploratory workflow,
- and clear direction for scalable production-oriented development.